Universidade do Minho
Departamento de Informática

Dados e Aprendizagem Automática
Ano letivo 2026/2027

Practical Exercise no. 3

Theme

Data Exploration and Preparation

Exercise

An in-depth exploration of the data enables conclusions to be drawn that are often hidden and may be important for understanding the domain and the problem at hand. In this practical exercise, a set of techniques will be applied to explore and process datasets.

Tasks

I - Sentiment Analysis Dataset

Download the dataset available at https://bit.ly/3ufnsgF, which contains data on users' sentiment towards a web platform.

Carry out the following tasks:

T1. Load the dataset. Apply methods for data exploration, i.e. analyze the data in relation to the users:

a. Central tendency;
b. Statistical dispersion;
c. Correlation between features.

T2. Create plots to visualize the data.

T3. Apply different data treatments, such as:

a. Treating missing values;
b. Removing duplicate records;
c. Create an *"age_binned"* attribute by creating three bins of equal frequency on the *"age"* feature;
d. For each record, extract the year, month and day of the week from the *"birthday"* feature;
e. Remove users from the platform who have an activity on the platform (*"WebActivity"*) of less than 1 hour and who are over 70 years old;
f. Use *"Sentiment Analysis"* to determine the mode of the remaining attributes;
g. Using *"Sentiment Analysis"* and *"MaritalStatus"*, obtain the average of the remaining attributes.

T4. Critically analyze the previous tasks. What conclusions can the company draw?

II - EPL Players Stats Dataset

Download the dataset, which is available at https://bit.ly/3525yDr, to access data on the performance of various football players in the 2017/2018 Premier League season.

Complete the following tasks:

T5. Load the dataset. Explore the data and look for relevant information to display.

T6. Critically analyze the dataset. For example, which team is the most undisciplined? What are the top 10 assists for goals? Which nationalities feature most prominently (top 5) in this edition of the Premier League?